Vertically Integrated Industries: Listening List by Industry (v2)

Compiled Oct 27, 2026, to match primer v9.3 (five industries). Every link was opened on Oct 27, 2026, loaded, and is free (YouTube, open podcast pages or free web pages). Lengths come from YouTube or the podcast host; "length not listed" means the page did not give one. ★ = top pick for the section. "Repo" gives the source ID. Company numbers are the companies' own claims unless marked otherwise.

Start here, if you only have 3 hours: a16z *China Has Mass* (macro) → a16z *Critical Minerals* (Raw Materials) → Relentless with Chris Power (Manufacturing) → YC on Solugen (Chemicals) → Latent Space with Radical AI (Advanced Materials).

0. Cross-cutting macro

1. ★ **"China Has Mass. Can America Catch Up?"**, a16z Podcast, Sep 29, 2025, length not listed. https://a16z.com/podcast/china-has-mass-can-america-catch-up/ — *Why:* Anduril's and Hadrian's CEOs explain the munitions gap: war games run out in about a week, restocking takes 2–3 years. Repo: src-20250929-452dcb.
2. **"How LA Is Becoming America's New Industrial Tech Hub"**, Bloomberg *Odd Lots*, Sep 21, 2026, 34 min. https://www.youtube.com/watch?v=mgCQ0Pl754c — *Why:* the best mainstream framing; the bottleneck is factory software and feedback, not robots. Repo: src-20260921-a51ab2.
3. **"Can the U.S. Still Build Things?" (#34)**, *The Drone Ultimatum*, with Miles Arnone (Re:Build), about 95 min. https://www.thedroneultimatum.com/can-the-us-still-build-things-fixing-the-industrial-base-miles-arnone-podcast-34/ — *Why:* the case for buying and upgrading old shops instead of building megafactories. Repo: src-20261022-f423ef.

1. Raw Materials (Minerals & Metals)

Stage A — Finding and mining

1. ★ **"Critical Minerals: Mining for the Industrial Future"**, a16z Podcast, Jul 23, 2025, length not listed. https://a16z.com/podcast/critical-minerals-mining-for-the-industrial-future/ — *Why:* Mariana's Turner Caldwell on owning mines and refineries and using software to cut build time and cost. Repo: src-20250723-3f3ef8.
2. **"Inside Earth AI: Cutting the Cost of Critical Mineral Discovery" (S2 E57)**, Rare Earth Exchanges, Feb 26, 2026, 33 min. https://www.youtube.com/watch?v=-dQONUziAxY — *Why:* founder Roman Teslyuk on training on 50 years of Australian drilling archives, and why few new deposits are found each year (his figures). Repo: src-20251001-cf441c.
3. **"From Y Combinator to mining, ft Earth AI's Roman Teslyuk"**, YouTube, Apr 28, 2026, 66 min. https://www.youtube.com/watch?v=KXbUJ2CjWbI — *Why:* the longer Earth AI founder story; pair with the primer's bear case (no resource estimate yet). **Not in the repo (new).**
4. **"Critical Minerals: What Makes Mining Tech VC-Backable"**, *Deep Tech Catalyst*, with Jun Ku (Main Sequence), length not listed. https://www.thescenarionist.com/p/critical-minerals-at-venture-scale-deeptech — *Why:* an investor on how non-dilutive government money makes heavy-capex mining fundable. Repo: src-20261022-cd6c5c.

Stage B — Refining and making metal

1. ★ **"How The Rarest Company On Earth Raised $75M"**, *Sourcery*, Aug 11, 2025, length not listed. https://podcasts.apple.com/us/podcast/how-the-rarest-company-on-earth-raised-%2475m/id1738124704?i=1000721503376 — *Why:* Vulcan's early demand: pilot plant opened Mar 2025 and was qualifying magnets with customers within four months (company claim). Repo: src-20250811-e7cb9a.
2. **"'Supply Chains Win Wars': Rare Earth Magnets with John Maslin"**, *Building the Base*, Apr 29, 2025, length not listed. https://buildingthebase.org/episode/supply-chains-win-wars-rare-earth-magnets-with-john-maslin-ceo-vulcan-elements — *Why:* the magnet macro in one sitting; the gap is processing and metal-making, not ore. Repo: src-20250429-6f9155.
3. **"Energy, Minerals, and the Physical Stack Behind AI"**, a16z Podcast, May 13, 2026, length not listed. https://a16z.com/podcast/energy-minerals-and-the-physical-stack-behind-ai/ — *Why:* Mariana's refinery and plant software (the data loop inside owned assets). Repo: src-20260513-bdc044.
4. **"Advancing US critical minerals with software and autonomy"**, Benchmark Mineral Intelligence, Jun 23, 2026, 23 min. https://www.youtube.com/watch?v=OvJX0j5OPXs — *Why:* how industry analysts see AI-run refineries. Repo: src-20260623-6dc76b.
5. *(Quick watch)* **"The Future is Metal – Mariana Minerals"**, a16z, Aug 5, 2026, 9 min. https://www.youtube.com/watch?v=uFt-tkagUUI — *Why:* nine minutes on China's 90%+ share of refining. Repo: src-20260805-b7e3c7.

2. Manufacturing

Metal Supply

1. ★ **"Just Gonna Send It Podcast: Zane Hengsperger (Nox Metals) – Ep. 13"**, SendCutSend, Mar 25, 2026, 54 min. https://www.youtube.com/watch?v=qhRV1cBG2lw — *Why:* Nox's founder in conversation with a big customer type; why software beats old-line distributors on quoting and cutting. Repo: src-20260325-5b68db.
2. **"Reindustrializing America: Zane Hengsperger on building Nox"**, *First Builders*, Apr 27, 2026, length not listed. https://podcasts.apple.com/us/podcast/reindustrializing-america-zane-hengsperger-on-building/id1818466924?i=1000763907900 — *Why:* the detailed founder story and lean-capital bet. Repo: src-20260427-83b102.
3. **"Building NOX Metals & The Future of Manufacturing" (#430)**, *Powers Pod*, length not listed. https://www.powerspod.com/reindustrializing-america-building-nox-metals-the-future-of-manufacturing-with-zane-hengsperger/ — *Why:* Nox as the AI-run metal service center. Repo: src-20261022-3135b5.

Fabrication

1. ★ **"How to Build a Manufacturing Company in America | Jim Belosic, SendCutSend"**, *The Peel*, Apr 30, 2026, 1 h 32 min. https://www.youtube.com/watch?v=Zajo84R1ckI — *Why:* a bootstrapped fabricator at about $140M run rate in eight years (company figure); the best repeat-order case study. Repo: src-20260430-9318cd.
2. **"What Everyone is Missing About Reindustrialization"**, Machina Labs, Oct 20, 2026, 55 min. https://www.youtube.com/watch?v=EiMhaHThSXk — *Why:* Machina's and Freeform's founders on why the money is in the finished part ($57 of steel becomes a $2,000 part). Repo: src-20261020-bac9ca.
3. **"The Crisis Facing American Manufacturing"**, *Core Memory* (Ashlee Vance), Sep 26, 2026, 25 min. https://www.youtube.com/watch?v=NpvYyx2OUTc — *Why:* a short documentary inside Hadrian and Machina Labs. Repo: src-20260926-70087f.

Precision & Defense Hardware

1. ★ **"How to Dominate Manufacturing in the US | Chris Power, Hadrian"**, *Relentless*, Sep 21, 2026, 1 h 19 min. https://www.youtube.com/watch?v=hmOLr_h7zTk — *Why:* the clearest unit-economics case for owning factories; about 80% of capex is shared across programs. Repo: src-20260921-876080.
2. **"From hypercars to cruise missiles: Lukas Czinger on the future of US defense" (E2292)**, *This Week in Startups*, May 23, 2026, 1 h 46 min. https://www.youtube.com/watch?v=U5kZI86lljk — *Why:* Divergent's CEO on moving from cars to missile airframes and why it builds its own printers. Repo: src-20260523-b05ded.
3. **"Building the Arsenal: Speed, Scale, and Affordability"**, Air & Space Forces Association, Oct 14, 2026, 40 min. https://www.youtube.com/watch?v=r7BNP3R_TWk — *Why:* the buyer side; Divergent's CEO on a panel about producing munitions at scale. Repo: src-20261014-47f64b.
4. *(Read or listen)* **TBPN transcript: Atomic Industries raises $25M Series A**, Sep 19, 2025. https://www.tbpndigest.com/story/2025-09-19/aaron-slodovs-atomic-industries-raises-25m-series-a-to-automate-injection-mold-manufacturing-in-detroit/transcript — *Why:* Atomic's move from software to molds to parts; "mid eight figure" contracts (company claim). Repo: src-20250919-2805c7.

3. Chemicals

1. ★ **"Inside The Startup Reinventing The $6 Trillion Chemical Manufacturing Industry"**, YC, Mar 2026, about 13 min. https://www.ycombinator.com/library/NO-inside-the-startup-reinventing-america-s-trillion-dollar-chemical-industry — *Why:* the best short intro to a fully integrated chemical maker (Solugen; ~96% yield claim). Repo: src-20260320-f90e80.
2. **"EP 312 Lee Cronin on Automating Chemistry"**, *The Jim Rutt Show*, Jul 2025, length not listed. https://jimrutt.substack.com/p/ep-312-lee-cronin-on-automating-chemistry-f9f — *Why:* Chemify's "code to molecule" and how it charges (company claims). Repo: src-20250724-0f42f3.
3. **"Why Twelve believes power-to-liquids will revolutionise SAF"**, *Sustainability in the Air*, Jul 24, 2025, 34 min. https://podcasts.apple.com/bm/podcast/why-twelve-believes-power-to-liquids-will-revolutionise/id1620281360?i=1000718768981 — *Why:* an owned e-fuels plant and its long-term buyers; policy is the swing factor. Not in the repo.
4. **"Mission-driven doctor, inventor & entrepreneur" (Ep. 175)**, *Molecule to Market*, Phlow CEO Eric Edwards, 55 min. https://moleculetomarketpod.com/episodes/mission-driven-doctor-inventor-entrepreneur/ — *Why:* drug-ingredient onshoring with the government as anchor buyer. Not in the repo.

4. Advanced Materials

Producers

1. ★ **"Jonathan Godwin, CEO at Orbital: Orbital raised $50m Series B"**, *Scaling Europe*, Jun 29, 2026, 21 min. https://open.spotify.com/episode/2iSBDNWM2Sc5czUAIW0ugX — *Why:* Orbital's founder on why it builds and sells hardware (a GPU-cooling material) instead of only models (company claims). Not in the repo (new).
2. **"Developing a supply chain for American-made batteries"**, *Volts*, Sep 6, 2024, length not listed (free transcript on page). https://www.volts.wtf/p/developing-a-supply-chain-for-american — *Why:* Mitra Chem's CEO on China's grip on LFP cathodes and its iron-rich roadmap. Not in the repo (new).

Materials Discovery Labs

1. ★ **"The Self-Driving Lab — Joseph Krause, Radical AI"**, *Latent Space*, Jun 17, 2026, length not listed. https://www.latent.space/p/radical-ai — *Why:* the moat is experiments, not models; customers asking for 300 lb pushed Radical toward making alloys. Repo: src-20260617-f9d293.
2. **"AI for Atoms: Periodic Labs with Liam Fedus"**, *No Priors*, Apr 3, 2026, 29 min. https://www.youtube.com/watch?v=Oru2Jxr1xHU — *Why:* economics of an AI lab running its own experiments. Repo: src-20260403-40fa0e.
3. **"Meet the Ex-OpenAI & DeepMind Leads Creating an AI Scientist"**, *The Generalist*, Sep 29, 2026, 60 min. https://www.youtube.com/watch?v=vJVZgeXKlVo — *Why:* Periodic's superconductor focus and the impurity problem. Repo: src-20260929-ca6d19.
4. **"Max Welling, CuspAI — Nebius Conversations"**, May 4, 2026, 19 min. https://www.youtube.com/watch?v=40CU4rbO7E4 — *Why:* the software-only model for contrast; it designs but does not make. Repo: src-20260504-36655f.

5. Enabling Software & Robotics (tracked, outside the core)

1. ★ **"Building Factory SuperIntelligence with Ariyan Kabir"**, *Machine Minds*, length not listed. https://machineminds.buzzsprout.com/2212148/episodes/19077384-building-factory-superintelligence-with-ariyan-kabir — *Why:* GrayMatter moved from licensing software to selling robot cells as a service, which tests the thesis. Repo: src-20261022-3335bd.
2. **"Former top OpenAI employees raise capital for defense startup valued at $1.5B"**, CNBC, Oct 16, 2026, short clip. https://www.cnbc.com/video/2026/10/16/former-top-openai-employees-raises-capital-for-defense-startup-valued-at-1point5b.html — *Why:* Arda on the 3–6 months it takes to retool factory automation. Repo: src-20261020-878aa9.
3. **"Mind Robotics: Building the AI-Native Robotics Platform"**, Kleiner Perkins, May 13, 2026, 4 min. https://www.youtube.com/watch?v=fc9GB9Yz75Y — *Why:* Rivian's line as Mind's live data loop. Repo: src-20260513-abe9cf.
4. **"Big Ideas 2026: Physical AI and the Industrial Stack"**, a16z Podcast, Dec 25, 2025, length not listed. https://a16z.com/podcast/big-ideas-2026-physical-ai-and-the-industrial-stack/ — *Why:* how investors frame the software layer going into 2026. Repo: src-20251225-725014.

- *Gap:* no free Lumaril podcast or founder interview found (searched Oct 27, 2026). Use the primer profile (section 7.10).